

MATHEMATICS FOR ML: STATISTICAL TESTS IN PRACTICE

Yogesh Haribhau Kulkarni

Outline

1 STATISTICAL TESTS IN PRACTICE

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- ▶ Feel free to follow me at:
 - ▶ Github (github.com/yogeshhk)
 - ▶ LinkedIn (www.linkedin.com/in/yogeshkulkarni/)
 - ▶ Medium (yogeshharibhaukulkarni.medium.com)
 - ▶ Send email to yogeshkulkarni at yahoo dot com



Office Hours:
Saturdays, 2 to 5pm
(IST); Free-Open to all;
email for appointment.

About Content

- ▶ Bare essential mathematics needed for Machine/Deep Learning
- ▶ Just to get you started.
- ▶ Each field within Machine/Deep Learning can go extremely deep.
- ▶ This is to give you enough foundation needed.

About Me

- ▶ I am not a mathematician.
- ▶ And this course won't turn YOU into one!!! (if you are not already)
- ▶ But, I will surely attempt to make you get interested in the learning, more.

Hypothesis Testing

Statistical Hypothesis Testing

- ▶ t-test = compares 1/2 numerical groups
- ▶ ANOVA = compares > 2 numerical groups
- ▶ Chi-square test = compares categorical variables

t-Test

t-Test

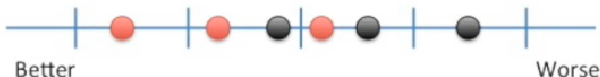
- ▶ There are two types of t-tests: paired and unpaired.
- ▶ Paired t-tests are used when you have 'before' and 'after' measurements from the same subjects. E.g body temperature before and after medicine.
- ▶ Unpaired: you are comparing height data for two groups.
- ▶ Subcategories of Unpaired t-tests:
 - ▶ Assumes that variations with both groups are same.
 - ▶ Does NOT Assume that variations with both groups are same.
- ▶ Another classification: 1-tail or 2-tails t-tests
- ▶ In case of comparing heights of two groups, the 2-tail t-test will evaluate if group A is Higher than B as well as Shorter than B, ie at both ends.
- ▶ It is used when you already know which direction has to be compared, higher or lower.

(Ref: StatQuest: Which t test to use? - Josh Starmer)

1 vs 2 tailed t-Tests

Example: a clinical trial is done on 6 patients to find effect of a new drug.

- ▶ Data shows that with new drug, results are much better (more red dots on left)
- ▶ The red dot in the middle is a bit problematic.
- ▶ After doing stats, you get p-value of 0.03 for 1 tail t test.
- ▶ 0.03 is smaller than threshold of 0.05 (CI). Meaning this new drug is significantly different. Its mean result is far away from other means found during existing treatments.
- ▶ 2-tailed gives p-value of 0.06. Not good.
- ▶ Which p-value to use?



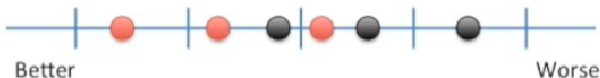
● = Your new treatment

● = The standard treatment

1 vs 2 tailed t-Tests

Example: a clinical trial is done on 6 patients to find effect of a new drug.

- ▶ 1 tailed p value tests the hypothesis that your new drug is better than the existing drug. Whereas,
- ▶ 2 tailed p-value tests whether the new drug is better , worse or not significantly different.
- ▶ 1 tailed p value is smaller because it does not distinguish between “worse” or “not significantly different”.
- ▶ AS we wish to know if the new drug was worse than the existing treatment, we should use 2 tailed p value test.



● = Your new treatment

● = The standard treatment

(Ref: StatQuest: One or Two Tailed P-Values - Josh Starmer)

t-tests

- ▶ One sample hypothesis testing
 - ▶ We are given population σ : use z distribution, z-test
 - ▶ We are Not given population σ , need to estimate it: t-distribution, t-test
- ▶ Two sample t-test

Z distribution

Same as Sampling distribution

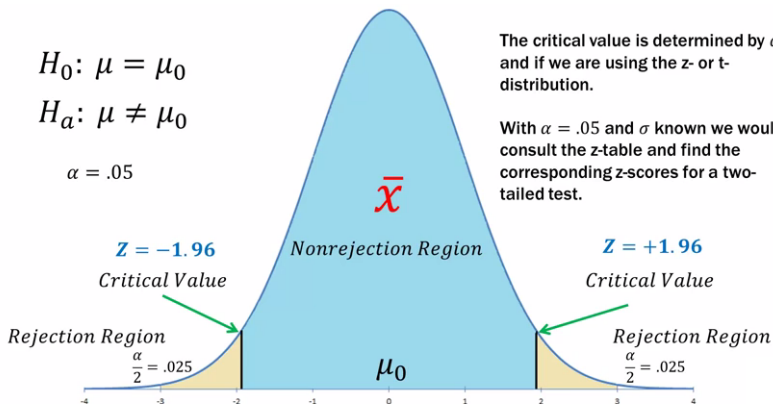
$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

$$\alpha = .05$$

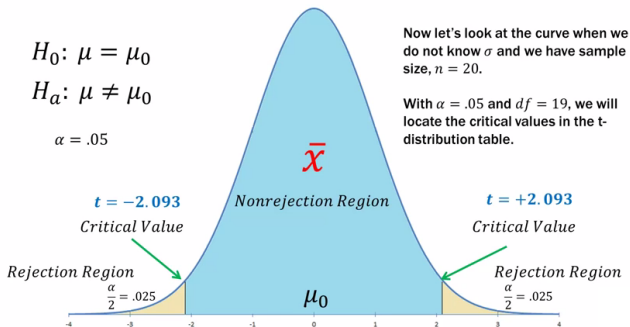
The critical value is determined by α and if we are using the z- or t-distribution.

With $\alpha = .05$ and σ known we would consult the z-table and find the corresponding z-scores for a two-tailed test.



t distribution

Every t distribution is unique and is specific to sample size. For $n=20$, look at t table with degrees of freedom $20-1=19$



Range is a bit more than Z dist. Like pushing it down from top!!

One Sample t-test

- ▶ Objective: Test if hypothesized population mean matches with the actual population mean.
- ▶ μ_0 is hypothesized population mean
- ▶ μ is the true/actual population mean
- ▶ Method: Test using sample means and confidence interval

One Sample t-test

- ▶ When?: To test whether a given sample is coming from a population whose mean is specified value

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- ▶ $\bar{x}$: sample mean of n observations
- ▶ s : sample std deviation of n observations
- ▶ μ_0 : hypothesized population mean
- ▶ $H_0 : \bar{x} = \mu_0$
- ▶ $H_1 : \bar{x} \neq \mu_0$

One Sample t-test

- ▶ t value computed is similar to z score.
- ▶ Shows how many std deviations you are away from mean.
- ▶ With the given t score, in the t table find the location on graph
- ▶ Looking at cut off boundaries, are you landing in the non rejection region or the rejection region?

Example: One Sample t-test

- ▶ New tyres launched are claim to have life of 40000 km.
- ▶ A retailer wants to test this claim.
- ▶ He has taken random sample of 8 tyres.
- ▶ He is testing life of the tyres under normal condition.

Example: One Sample t-test

- ▶ We don't know population std dev. So t test.
- ▶ $H_0; \mu_0 = 40000$
- ▶ $H_1; \mu_0 \neq 40000$
- ▶ Lets take $\alpha = 0.05$
- ▶ $df = n - 1 = 8 - 1 = 7$

Example: One Sample t-test

t Table

df		Critical Values			
1-tailed	0.1	0.05	0.025	0.01	0.005
2-tailed	0.2	0.1	0.05	0.02	0.01
1	3.0777	6.3138	12.7062	31.8208	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123
14	1.3450	1.7613	2.1448	2.6245	2.9768
15	1.3406	1.7531	2.1314	2.6025	2.9467
16	1.3368	1.7469	2.1199	2.5835	2.9208
17	1.3334	1.7396	2.1098	2.5669	2.8982
18	1.3304	1.7341	2.1009	2.5524	2.8784
19	1.3277	1.7291	2.0930	2.5395	2.8609
20	1.3253	1.7247	2.0860	2.5280	2.8453

- ▶ Critical boundary-value = 2.3646
- ▶ If t false below -2.3646 or above 2.3646, then reject H_0 .

Example: One Sample t-test

- His findings below:

Sr	Km
1	35000
2	38000
3	42000
4	41000
5	39000
6	41500
7	43000
8	38500

- Given goal mean $\mu_0 = 40000$
- $n = 8$
- Calculate sample mean $\bar{x}$, it comes to 39750
- Calculate s , it comes to 2618.615
- $df = n - 1 = 7$

Example: One Sample t-test

- ▶ $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{39750 - 40000}{2618.615/\sqrt{8}} = \frac{-250}{925.82} = -0.27$
- ▶ It is right of negative boundary and left of positive boundary,
- ▶ So, in valid region.
- ▶ So, cannot reject H_0 .
- ▶ So, no difference.
- ▶ Sample mean is same as given mean.

Two Samples t-test

- ▶ When?: To test whether given two samples are coming from a population

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p / \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- ▶ $\bar{x}_1$: sample mean of n_1 observations
- ▶ $\bar{x}_2$: sample mean of n_2 observations
- ▶ s_p : Pooled std deviation of two samples
- ▶ μ_0 : specified population mean
- ▶ $H_0 : \bar{x}_1 = \bar{x}_2$
- ▶ $H_1 : \bar{x}_1 \neq \bar{x}_2$

Example: Two Sample t-test

- ▶ Is there any appreciable difference of IQs of Males and Females?
- ▶ $H_0 : x_{male}^- = x_{female}^-$
- ▶ $H_1 : x_{male}^- \neq x_{female}^-$
- ▶ Lets take $\alpha = 0.05$
- ▶ $n_m = n_f = 18; n = 36$
- ▶ $x_{male}^- = 98.9$
- ▶ $x_{female}^- = 102.4$
- ▶ $s = 12$; same for both

Example: Two Sample t-test

- ▶ Finding critical values value for t at $1 - \alpha/2$ for two tail test and $df = n - 1 = 35 = 2.030$
- ▶ $t = \frac{\bar{x}_{female} - \bar{x}_{male}}{s_p / \sqrt{\frac{1}{n_f} + \frac{1}{n_m}}}$
- ▶ $= \frac{102.4 - 98.9}{12 / \sqrt{1/18 + 1/18}} = 0.097$
- ▶ Lies in acceptable region.
- ▶ Can not reject H_0
- ▶ No diff in IQs.

Two Samples t-test

What is t-test?

- ▶ Compares two averages (means) and tells you if they are different from each other.
- ▶ Tells you how significant the differences are
- ▶ It lets you know if those differences could have happened by chance.

Two Samples t-test

What is t-test?

- ▶ The t score is a ratio between the difference between two groups and the difference within the groups.
- ▶ A large t-score tells you that the groups are different.
- ▶ A small t-score tells you that the groups are similar.
- ▶ The bigger the t-value, the more likely it is that the results are repeatable.

Two Samples t-test

- ▶ Every t-value has a p-value to go with it. A p-value is the probability that the results from your sample data occurred by chance.
- ▶ a p-value of .01 means there is only a 1% probability that the results from an experiment happened by chance.

Two Samples t-test

Types of t-tests?

- ▶ An Independent Samples t-test compares the means for two groups.
- ▶ A Paired sample t-test compares means from the same group at different times (say, one year apart).
- ▶ A One sample t-test tests the mean of a single group against a known mean.

Two Samples t-test

Example: to test whether the height of men in the population is different from height of women in general.

Steps:

- ▶ Determine a null and alternate hypothesis.
 - ▶ Null-Hypothesis: no statistically significant difference, height of men and women are the same
 - ▶ Alternate-Hypothesis: statistically significant difference, height of men and women are different
- ▶ Collect sample data. Two sets: men and women. Generally sizes should be same, but can be different, n_x and n_y

Two Samples t-test

Steps (cont.):

- ▶ Determine a confidence interval and degrees of freedom
- ▶ α of 0.005 is 95% confidence that the conclusion of this test will be valid
- ▶ The degree of freedom $df = n_x + n_y - 2$
- ▶ Calculate the t-statistic

$$t = \frac{M_x - M_y}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}}$$

$M = \text{mean}$
 $n = \text{number of scores per group}$

$$S^2 = \frac{\sum (x - M)^2}{n - 1}$$

$x = \text{individual scores}$
 $M = \text{mean}$
 $n = \text{number of scores in group}$

Two Samples t-test

Steps (cont.):

- Calculate the critical t-value from the t distribution

cum. prob one-tail two-tails	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
df	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.316
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.516	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.058	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	96%	99%	99.8%	99.9%
	Confidence Level										

- or call ready Python function

Two Samples t-test

Generate data

```
1 #Sample Size
  N = 10
3 #Gaussian distributed data with mean = 2 and var = 1
  a = np.random.randn(N) + 2
5 #Gaussian distributed data with with mean = 0 and var = 1
  b = np.random.randn(N)
```

Two Samples t-test

```
1 #Calculate the variance to get the standard deviation
2 #For unbiased max likelihood estimate we have to divide the var by N-1, and
   therefore the parameter ddof = 1
3
4 var_a = a.var(ddof=1)
   var_b = b.var(ddof=1)
5
6 #std deviation
7
8 s = np.sqrt((var_a + var_b)/2)
```

Two Samples t-test

```
## Calculate the t-statistics
2 t = (a.mean() - b.mean())/(s*np.sqrt(2/N))
## Compare with the critical t-value
4 #Degrees of freedom
df = 2*N - 2
6 #p-value after comparison with the t
p = 1 - stats.t.cdf(t,df=df)
8 #Note that we multiply the p value by 2 because its a twp tail t-test

10 ## Cross Checking with the internal scipy function
t2, p2 = stats.ttest_ind(a,b)
```

Paired t-test

- ▶ When?: To check, effectiveness of a new treatment, a new method employed
- ▶ observations on every unit are made before and after applying the treatment or method.
- ▶ Hence if treatment or method is effective, there will be significant difference in observations before applying it and after applying it.

$$t = \frac{\bar{x}_D - \mu_0}{\sigma_D / \sqrt{n}}$$

- ▶ $\bar{x}_D$: mean difference of n paired observations
- ▶ σ_D : std deviation of difference of n paired observations
- ▶ μ_0 : mean difference of n paired observations under H_0
- ▶ H_0 : Mean of samples before and after treatment is same.
- ▶ H_1 : Mean of samples before and after treatment is not same.

Chi-Square test

- ▶ When?: To test whether any two categorical variables are associated with each other or they are independent of each other,

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- ▶ O_i : Observed frequency of i th variable
- ▶ E_i : Expected frequency of i th variable
- ▶ H_0 : Two variables are independent of each other.
- ▶ H_1 : Two variables are not independent of each other.

F test

- ▶ When?: to know whether these two sample have same population variance

$$F = \frac{s_1^2}{s_2^2}$$

- ▶ S_1^2 Sample variance of first sample of size n_1
- ▶ S_2^2 Sample variance of second sample of size n_2
- ▶ H_0 : Variance ratio is one
- ▶ H_1 : Variance ratio not equal to one

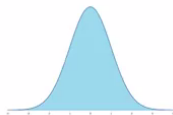
Anova

Analysis of Variance (ANOVA)

The difference between ANOVA and the t tests is that ANOVA can be used in situations where there are two or more means being compared, whereas the t tests are limited to situations where only two means are involved.

Analysis of Variance (ANOVA)

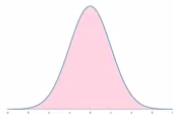
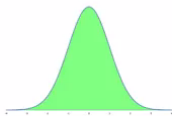
Suppose we want to compare **THREE** sample means to see if a difference exists somewhere among them.

 $\bar{x}_1$

Is one mean so far away from the other two that it is likely not from the same population?

What we are asking is:

Do all three of these means come from a common population?

 $\bar{x}_2$  $\bar{x}_3$

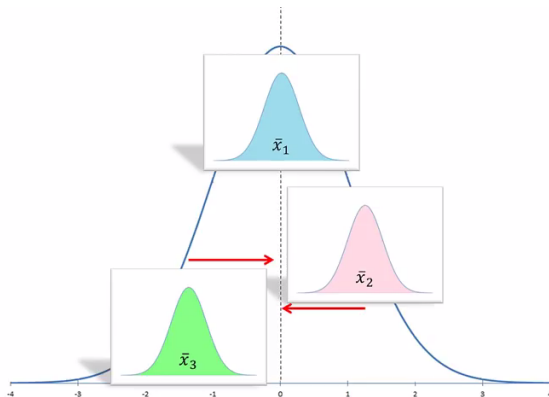
Or are all three so far apart that they **ALL** likely come from unique populations?

Finding **DISTANCE** between means

(Reference: Statistics 101 - Brandon Foltz)

Analysis of Variance (ANOVA)

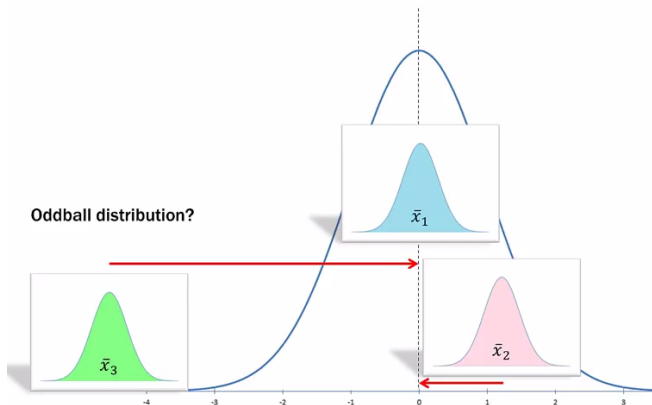
Combine all data points from all 3 samples. Draw its distribution. Then compare of each with the combined one.



$\bar{x}_1$ bang on combined mean, rest are a bit away.

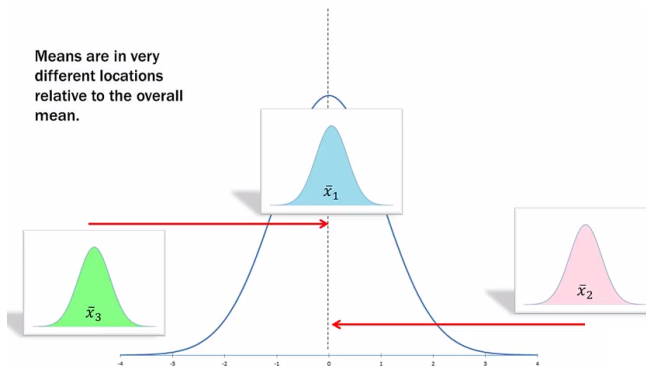
Analysis of Variance (ANOVA)

New example: where $\bar{x}_3$ is far away. So not part of combined population.
Outlier.

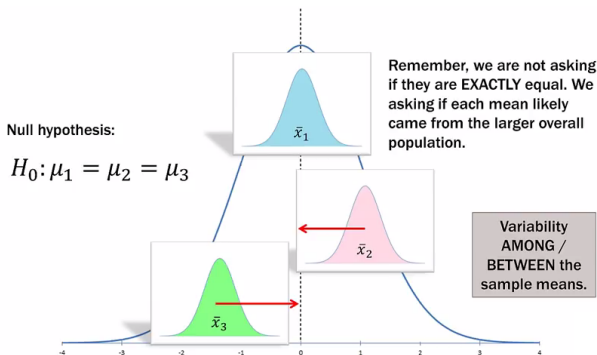


Analysis of Variance (ANOVA)

New example: where both $\bar{x}_2$ and $\bar{x}_3$ are far away. All three are their OWN populations.



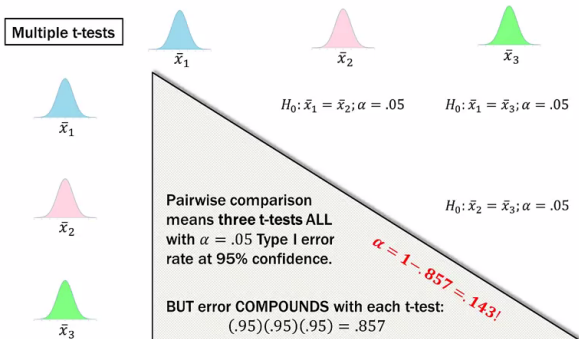
Analysis of Variance (ANOVA)



Variability is between means.

Analysis of Variance (ANOVA)

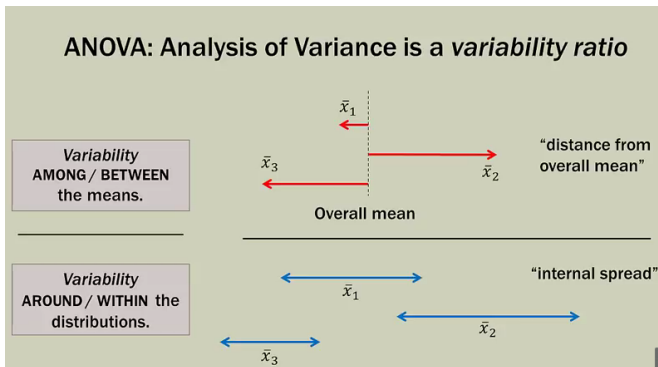
Pair wise t tests



Combined error rate went from 5% to 14.3%. So we do not conduct multiple t tests.

Analysis of Variance (ANOVA)

Each distribution has its own internal variance / variability. Anova is a variability ratio.



Analysis of Variance (ANOVA)

Total variance has two components.

ANOVA: Analysis of Variance is a *variability ratio*

$$\frac{\text{Variance Between}}{\text{Variance Within}} \left. \vphantom{\frac{\text{Variance Between}}{\text{Variance Within}}} \right\} \text{Total Variance Components}$$

$$\text{Variance Between} + \text{Variance Within} = \text{Total Variance}$$

“Partitioning” – separating total variance into its component parts

If the variability **BETWEEN** the means (distance from overall mean) in the numerator is relatively large compared to the variance **WITHIN** the samples (internal spread) in the denominator, the ratio will be much larger than 1. The samples then most likely do **NOT** come from a common population; **REJECT NULL HYPOTHESIS** that means are equal.

Analysis of Variance (ANOVA)

ANOVA: Analysis of Variance is a *variability ratio*

$$\frac{LARGE}{small} = \text{Reject } H_0$$

At least one mean is an outlier and each distribution is narrow; distinct from each other.

Variance Between
Variance Within

$$\frac{similar}{similar} = \text{Fail to Reject } H_0$$

Means are fairly close to overall mean and/or distributions overlap a bit; hard to distinguish.

$$\frac{small}{LARGE} = \text{Fail to Reject } H_0$$

The means are very close to overall mean and/or distributions "melt" together.

Analysis of Variance (ANOVA)

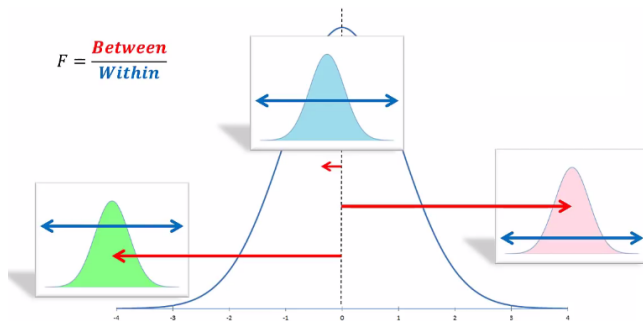
		
Year 1 Scores	Year 2 Scores	Year 3 Scores
82	71	64
93	62	73
61	85	87
74	94	91
69	78	56
70	66	78
53	71	87
$\bar{x}_1 = 71.71$	$\bar{x}_2 = 75.29$	$\bar{x}_3 = 76.57$

Overall Mean:

The mean of all 21 scores taken together.

$$\bar{\bar{x}} = 74.52$$

Analysis of Variance (ANOVA)



Thanks ...

- ▶ Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
- ▶ Call + 9 1 9 8 9 0 2 5 1 4 0 6



(<https://www.linkedin.com/in/yogeshkulkarni/>)



(<https://medium.com/@yogeshharibhaukulkarni>)



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